

Hariharan Narlakanti

AI Engineer — Backend Systems — GenAI

+91 8686937565 | narlakantihariharan@gmail.com | Hyderabad, India | LinkedIn | GitHub

SUMMARY

Computer Science graduate with strong fundamentals in Data Structures & Algorithms, OOP, DBMS, and backend software engineering, with experience building AI-powered systems using Python, FastAPI, Docker, and LLM pipelines. Developed production-style GenAI applications involving document ingestion, NLP inference, prompt orchestration, and scalable AI workflows, with applied knowledge of Retrieval-Augmented Generation (RAG), embeddings, and AI agents.

EDUCATION

Malla Reddy University, Hyderabad

2022 – 2026

Bachelor of Technology – Computer Science & Engineering (AI & ML), CGPA: **8.89 / 10**

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, OOP

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL

Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST API Design

Databases: PostgreSQL, MongoDB, MySQL

GenAI / AI Systems: LLM APIs, RAG, Prompt Engineering, Embeddings, AI Agents, Transformers, NLP

ML Frameworks: PyTorch, TensorFlow, CNNs

DevOps & Tools: Docker, Git, Linux, Azure AI Foundry, Render, Vercel, PyMuPDF

PROJECTS

Saaram — AI-Powered Telugu Summarization Platform

GitHub | Live Demo

- Designed and deployed a FastAPI-based AI inference platform serving **5+** REST endpoints with dynamic model routing, fallback handling, and latency-aware inference optimization.
- Built and containerized a full-stack system (React/Vite frontend, FastAPI backend) deployed across cloud platforms using Docker and lazy model loading under constrained compute.
- Benchmarked TF-IDF, mT5 Base, and fine-tuned mT5 models across a **1,302**-sample dataset to optimize latency-quality tradeoffs for production deployment.

ResearchCompass — AI Research Review Platform

GitHub

- Built a FastAPI backend that parses uploaded PDF research papers using PyMuPDF and generates structured multi-section reviews using LLM APIs via Azure AI Foundry.
- Designed a document intelligence pipeline for extraction, context retrieval, structured reasoning, and AI-powered response generation.
- Refactored an initial multi-agent workflow into a streamlined inference pipeline, improving output reliability and reducing orchestration complexity.

AgroAid — AI-Based Crop Disease Detection

GitHub

- Built a Flask-based inference service integrating a pretrained PyTorch CNN with a structured recommendation pipeline for crop disease prediction across **39** categories.
- Delivered multilingual treatment recommendations in English, Hindi, and Telugu using translation APIs and AJAX-based request handling.

ACHIEVEMENTS

- Best Project Award, Malla Reddy University – Telugu NLP capstone project.
- Co-authored a research paper on Telugu text summarization, submitted to CIS 2026 (Springer LNNS).